Second Assignment (Event Detection and Video Summarization)

TUTORS: MIRELA POPA

TEACHING ASSISTANT: AASHUTOSH GANESH

DEADLINE: 14/05/2026 AT 11.59 PM

DESCRIPTION

In this assignment, you will develop a video summary by selecting and combining important segments (events) from a video. These events should reflect the key moments that occur throughout the video in a logical sequence. The assignment consists of: annotating videos, detecting events using Vision-Language Models (VLMs), retrieving temporal segments using moment retrieval models, evaluating alignment with annotations and generating and analyzing video summaries. Your findings will be presented in a report and short presentation.

Methodology

1. **Annotating Videos**.
   For each video, you must annotate a set of events following the guidelines provided in the annex. After watching the video decide which are the key events which you would like to see in a summary. For each event, you should provide a description of what happens, the time interval in which it occurs and the subjectivity score, indicating how essential the event is to understand the video.
2. **Pipeline for Video Summarization**
   The main goal of the assignment is to construct a pipeline which begins by annotating videos, where key events are identified, described, and assigned timestamps and subjectivity scores. Next, a vision-language model (VLM) is used to automatically detect events from the video, either as a sequence of descriptions or with associated timestamps. These detected events are then used as queries for moment retrieval models (e.g., Moment-DETR), which localize the precise temporal segments where each event occurs. The retrieved segments are evaluated against the annotations, refined if necessary, and finally combined into a coherent video summary that reflects the most important events in order.
3. **Event detection using Vision Language models (VLMs)**
   Given the video, utilize a small scale vision language model (e.g., https://huggingface.co/blog/smolvlm2, or https://huggingface.co/Qwen/Qwen2.5-VL-3B-

Instruct-AWQ) to detect the events present in the video, in the following ways: a) event only: given the input video, the output consists of a list of detected events (e.g., person enters room, sits down, ..) and b) event + time-stamp: given the input video, the pre-trained vision-language model (VLM) provides an ordered list of detected events with temporal localization (e.g., person enters room, 00:03–00:07). Choose a VLM which can either ingest a video or a sequence of sampled frames. Design a prompt for the sampled frames/videos to do the following: "Retrieve all of the salient events of the video". Ensure the format of the events are listed in the following sequential order: (Salient event 1, Salient event 2, etc.). With the outputs provided by the VLM, write a parser that extracts the events created by the VLM. Using your annotated list of events, check if the salient events are present in the outputs generated by the VLM. This could be achieved by comparing the word embedding similarity between the parsed events captured by the vision language model and the annotated events. For the evaluation phase conduct a failure analysis to examine the types of events that were missed out by the models of your choosing. Alongside this, document your history of prompts and the success/failures of different prompting strategies in retrieving the event predictions.

4. **Moment-retrieval using pre-trained models**
   Use detected events (from step 3) as queries, apply two moment retrieval models (e.g., https://github.com/jayleicn/moment_detr, https://github.com/wjun0830/QD-DETR, https://github.com/line/lighthouse, or https://github.com/wjun0830/CGDETR) to retrieve relevant video segments and predict timestamps for each event (please check the provided code in Canvas for getting you started on this task). You should also experiment with different video features. For example, Moment-DETR expects pre-extracted features, not a raw video. Therefore, a typical pipeline includes sampling frames or clips, using a pretrained backbone (e.g., CLIP / SlowFast) and extracts features per time step. Alternatively, you can check the HERO feature extraction technique (https://github.com/linjieli222/HERO_Video_Feature_Extractor). After preparing your video features, feed them and a text query (event) and you should get the timestamps and the associated confidence score. As Moment-Retrieval relies on text-video alignment, you could also test multiple phrasings per event (e.g. 3-5 variations for each text query), run Moment-DETR for each and combine results, using max scoring or voting. In case the provided outputs are noisy, you could apply post-processing by merging overlapping segments, removing very short segments and applying temporal consistency rules.
5. **Evaluate the accuracy of the moment retrieval boundaries**
   Using your annotations and the retrieved moments/events, compute the IoU metric and conduct a failure analysis of various modeling strategies. Specify the types of queries the model fails to detect. (Optional - develop strategies that improve the performance of some of the chosen models, as described above and compare the improvements using either the IoU or a metric proposed by you).
6. **Video Summary**
   In the final step, you transform the retrieved events into a coherent and concise video summary. Using the timestamps predicted by your moment retrieval model, you extract the corresponding video segments for each selected event. The selected clips are concatenated to form a single summary video. You may reason about the summary characteristics by reflecting on: a) Relevance: Does each segment contribute to understanding the main message? b) Coverage: Are all important events included? c) Conciseness: Is unnecessary content removed? and d) Coherence: Does the summary flow logically?

Even though you will work with a limited amount of videos, which cannot provide statistical significant results, the main goals of the assignment are to understand event representation and learn how to define and annotate meaningful events in videos; leverage modern models, by applying vision-language models and moment retrieval methods in a zero-shot setting; evaluate model performance, by comparing predicted events and timestamps against ground truth annotations; analyze subjectivity, by reflecting on how event selection impacts the final summary and finally build a complete pipeline, by integrating detection, retrieval, and summarization into a working system.

DELIVERABLES:

1. Minimal requirements for passing the assignment: you should deliver a presentation on the 11th of May and the annotations of the videos assigned to your group.
2. Presentation of 5-7 minutes in which you should discuss your main findings, challenges, observations, ideas for improvement and critical reflections. Consider what it could be interesting to share with the other groups, given your main achievements. Please include one complete example from annotations to video summary.
3. A report of max. 1500-2000 words, including examples and critical reflections over the various steps of the assignment. Do not forget to thoroughly discuss your main results for different conditions and various algorithms/frameworks. Provide your evaluation procedure, discussing how your annotations perform for various design choices, prompts, moment retrieval frameworks (compare at least two options for steps 3 and 4).
4. Well documented code in python (a script or a jupyter notebook).
5. Annotations – please use the provided guidelines.
6. Upload everything on Canvas as a zip file, using the following format: Group Number_Second_Assignment_CV2026.zip.

GRADING

A) CODE – 0.75 POINTS

(CRITERIA: YOUR OWN PIPELINE, RUNNING, COMPLETE, BY INCLUDING ALL THE STEPS AND COMMENTED)

B) REPORT– 0.75 POINT

(IT IS IMPORTANT TO INCLUDE A CRITICAL DISCUSSION OF THE MAIN STEPS AND YOUR REFLECTIONS ON THE PROCESS, HOW TO TACKLE MAIN ENCOUNTERED CHALLENGES, DISCUSSION AND ANALYSIS OF THE MAIN DECISIONS, ESPECIALLY HOW THEY IMPACT THE DIFFERENCES BETWEEN YOUR ANNOTATIONS AND THE OBTAINED PREDICTIONS)

C) ANNOTATIONS – 0.5 POINTS

(COMPLETE FOR THE VIDEOS ASSIGNED TO YOUR GROUP, USING THE PROVIDED GUIDELINES)

D) PRESENTATION – 0.5 POINTS

(CRITICALLY REFLECT ON YOUR PIPELINE AND ANALYSIS RESULTS, REPORT ANY OBSERVATIONS AND FINDINGS, PROVIDE ONE EXAMPLE SUMMARY)

Appendix:

1. You can download the corresponding videos for your group from: https://surfdrive.surf.nl/s/DAS4bz2ZKj3JY2Z

Student Group	Dataset	Video Numbers
1	TVSUM	1,2,3,4
2	TVSUM	5,6,7,8
3	TVSUM	9,10,11,12
4	TVSUM	13,14,15,16
5	TVSUM	17,18,19,20
6	TVSUM	21,22,23,24
7	TVSUM	25,26,27,28
8	TVSUM	29,30,31,32
9	TVSUM	33,34,35,36
10	TVSUM	37,38,39,40
11	TVSUM	41,42,43,44
12	TVSUM	45,46,47,48
13	TVSUM and SumME	49,50,1,2
14	SumME	3,4,5,6
15	SumME	7,8,9,10,11
16	SumME	12,13,14,15
17	SumME	16,17,18,19
18	SumME	20,21,22,23
19	SumME and TVSUM	24,25,1,2,3

2. **Annotation Procedure**

For each video, you must annotate a set of events following the guidelines below.

Annotation requirements: please provide a **minimum of 4 events and a maximum of 16 salient events** per video. You may include more than 16 events if necessary, depending on the video length.

Each event included in your summary should be chosen based on its objective relevance to the overall message of the video. Since the selection of events in a video can be subjective, you are asked to reflect on how objective or subjective your chosen segments are. Assume that your goal is to select events that allow a viewer to fully understand the underlying message of the video. Based on this, justify your selection of segments and discuss any subjectivity in your decisions.

For each video provide:

a. A salient event is defined as an event that is important for understanding the overall video. Each event should: be clearly distinguishable from other events, be selected based on relevance, distinctiveness, and impact, be written in 1–2 concise sentences (avoid unnecessary detail), represent the event using as few frames as possible (i.e., keep it

temporally precise), be listed in chronological order and be derived from visual content, spoken dialogue or audio cues.

b. Time-stamps: Format: **MM:SS - MM:SS;** each time-stamp should include only the essential portion of the video needed to capture the event and avoid irrelevant or extra footage, while accurately matching the described event.

c. Subjectivity rating: Since the selection of events in a video can be subjective, you are required to reflect on your choices when constructing the summary. Specifically, you should distinguish between *objectively necessary events (*events that are essential for understanding the main message or narrative of the video) and *subjectively selected events (*additional events that you chose to include because you consider them important or relevant, even if they are not strictly required for understanding the overall narrative). Based on this distinction, justify your selection of events and discuss how subjectivity influenced your summary.

RATING	DESCRIPTION
2	An objectively essential event that is critical for understanding the main message, narrative, or content of the video
1	An objectively important event that contributes to understanding the video, but is not strictly critical
0	An event whose status is uncertain (objective or subjective), but is still considered necessary for constructing a coherent summary
-1	A subjective event that is somewhat helpful or provides minor additional context
-2	A subjective event that significantly enhances the summary, but is not required to understand the main message

Key principles to follow: be concise but informative, focus on important, distinct events, avoid redundancy or overlap between events, ensure timestamps are precise and apply consistent judgment when assigning subjectivity ratings.

Each video should be annotated independently by two annotators. The agreement between the annotations should be quantified using an inter-annotator agreement metric (e.g., Cohen's Kappa), considering the consistency in selected events, their temporal boundaries, and their assigned saliency ratings.

Guidelines: Annotate the events in the following format: Event description, time step (formatted as MM:SS -MM:SS), subjective score rating. For each video assigned to you, please save the annotations as text files in the following format [GroupID]-[Dataset]-[Video_name]-[StudentID].txt (to help distinguish between the two sets of annotations for each video)

3. Example Annotations

TVSUM Video 18

Event Description, Time-stamps, Subjectivity rating

- The host introduces the concept of "poor man's meals", 00:03 - 00:06, 2
- The host describes the meal that will be prepared in the video, 00:46 - 00:50, 2
- The host shows the ingredients and discusses the price of each, 00:52 - 01:00, 2
- The host sharpens a knife to prepare for cutting vegetables, 01:05 - 01:08, 1
- The host makes a joke about eating onions, 01:39 - 01:45, -2

- The host cuts onions using the correct technique, 02:15 - 02:27, 2
- The host cuts bell peppers, 02:37 - 02:57, 2
- The host shows the sausages used for the dish, 03:03 - 03:10, 2
- The host adds ketchup to caramelize for the dish, 03:56 - 04:01, 2
- The host organizes the ingredients for cooking, 04:15 - 04:28, 1
- The host instructs to cook the dish for 45 minutes and transitions to the final shot, 04:42 - 04:55, 1
- The host eats the food with bread and presents it as a serving suggestion, 06:05 - 06:20, 1

SumMe Video 12

Event Description, Time-stamps, Subjectivity rating

- Two individuals pour water onto what appears to be a slide, 00:00 - 00:03, 2
- A man starts sliding down a very large slide, 00:08 - 00:12, 2
- A man lands into a large bucket of water, 00:15-00:16, 2
- The man stands in the bucket while others celebrate, 00:21 - 00:23,1
- The crowd cheers for the man, 00:29- 00:31, -1